Assignment 1

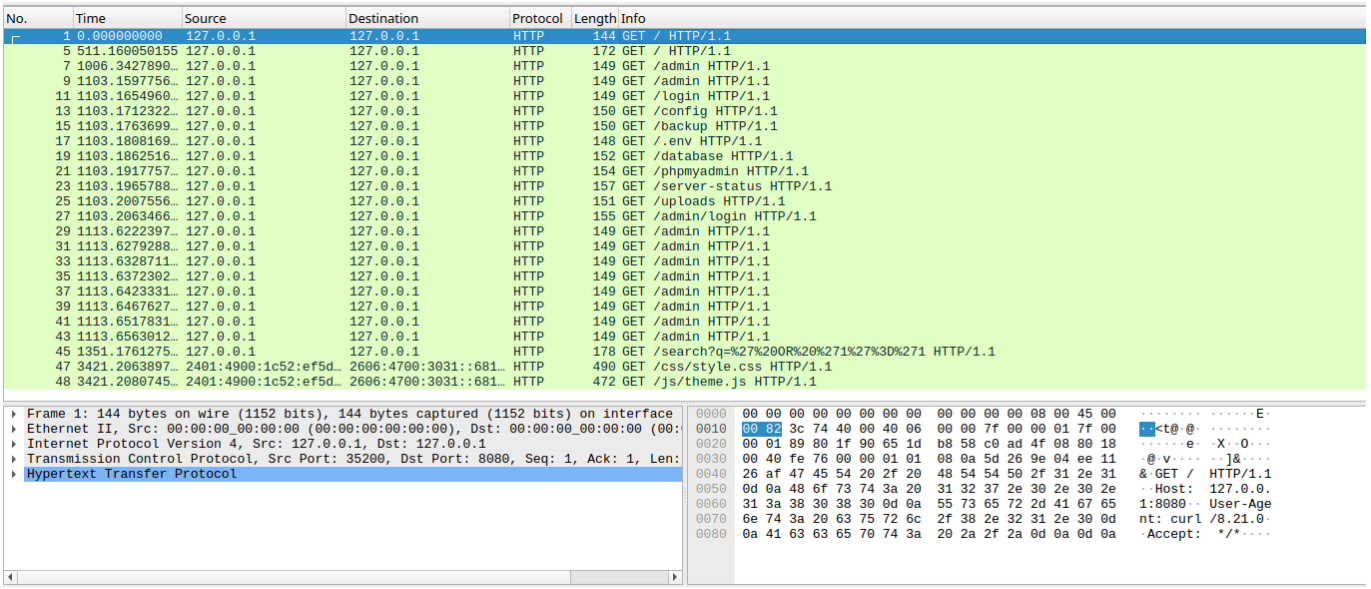
```
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Course : MCA
Subject : Data Communications and Computer Networks
```

1 Wireshark HTTP Traffic Analysis

This exercise focuses exclusively on the analyzing HTTP packets using Wireshark. A pre- captured tracefile.pcapng file is provided to you. No additional traffic generation is required. The objective is to

Task 1: Analyze a GET Request and Response

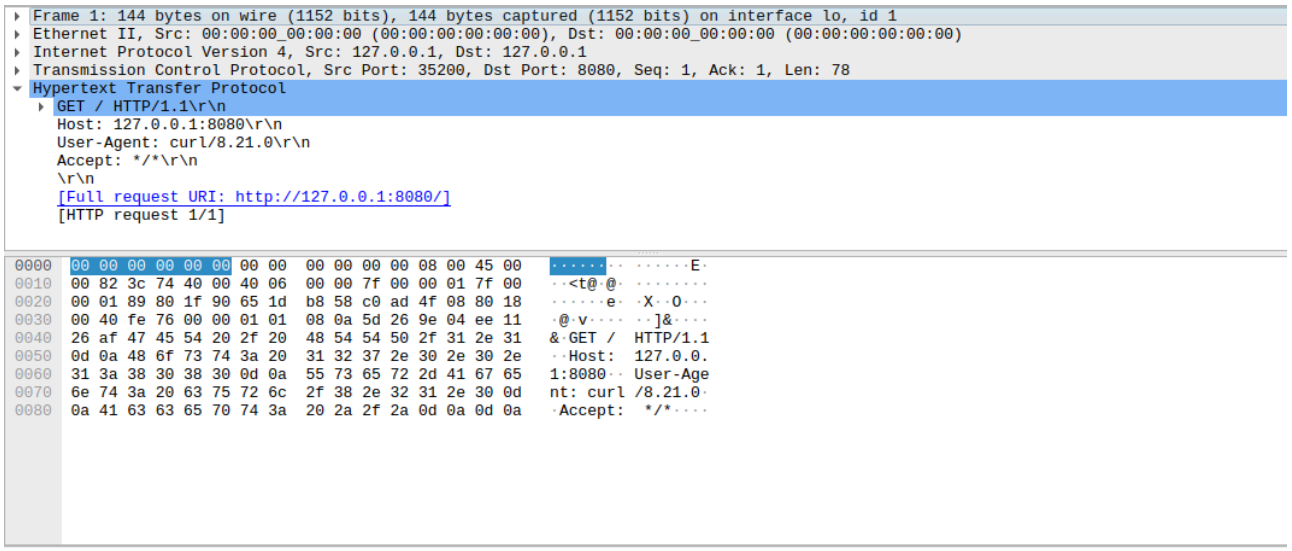
Identify a GET request in the provided capture and analyze its corresponding HTTP response. http.request.method == "GET"



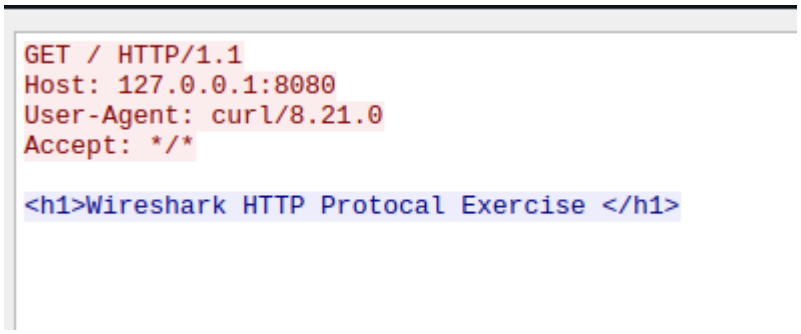
Questions

1. Identify one GET request and record its frame number.

```
request has Frame 1
```



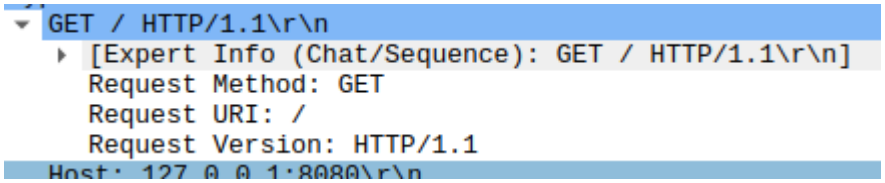
2. Identify the corresponding HTTP response and record its frame number.




```
frame no. 2
```

3. What is the requested URI?

```
Requested URI : /
```



4. What is the HTTP version?

```
HTTP/1.1
```

5. What is the value of the Host field?

```
Host : 127.0.0.1:8080
```

6. What is the reported User-Agent?

```
User-Agent curl/8.21.0
```

7. What HTTP response status code was returned?

```
GET / HTTP/1.1
Host: 127.0.0.1:8080
User-Agent: curl/8.21.0
Accept: */*

<h1>Wireshark HTTP Protocal Exercise </h1>
```

```
no response code returned
```

8. What does the response status code indicate?

```
N/A
```

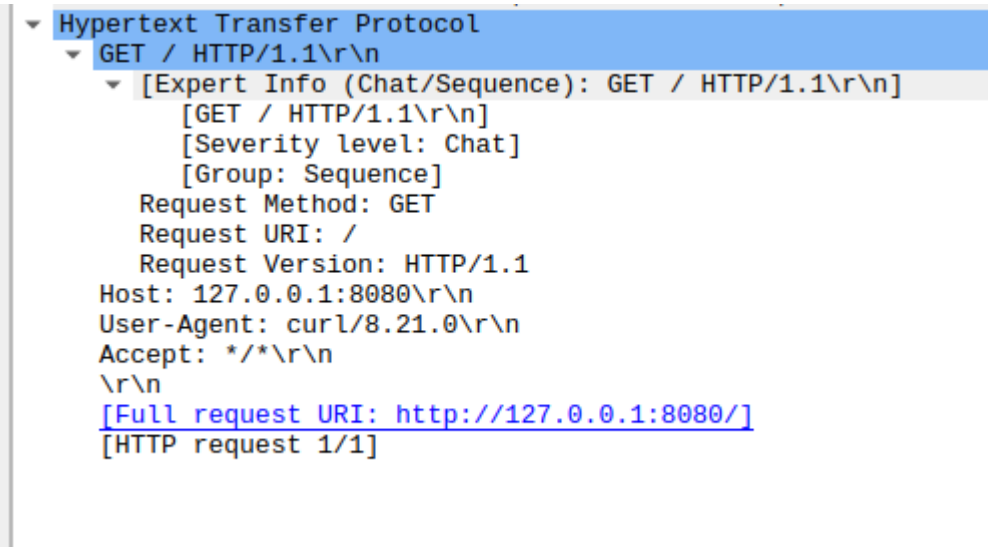
9. What is the response Content-Type?

```
Not present
```

10. Does the response contain a response body?

```
Yes
<h1>Wireshark HTTP Protocal Exercise </h1>
```

11. Identify three HTTP header fields and explain their purpose.




```
Request method : tells what kind of request was sent (GET : to get
some resource)
Request URI : to which uri path was the request sent
Request Version : what is the version of http is used to send request
```

Task 2: Analyze a POST Request and Response

Identify a POST request and examine how the request differs from the GET requests observed in the previous task. http.request.method == "POST"

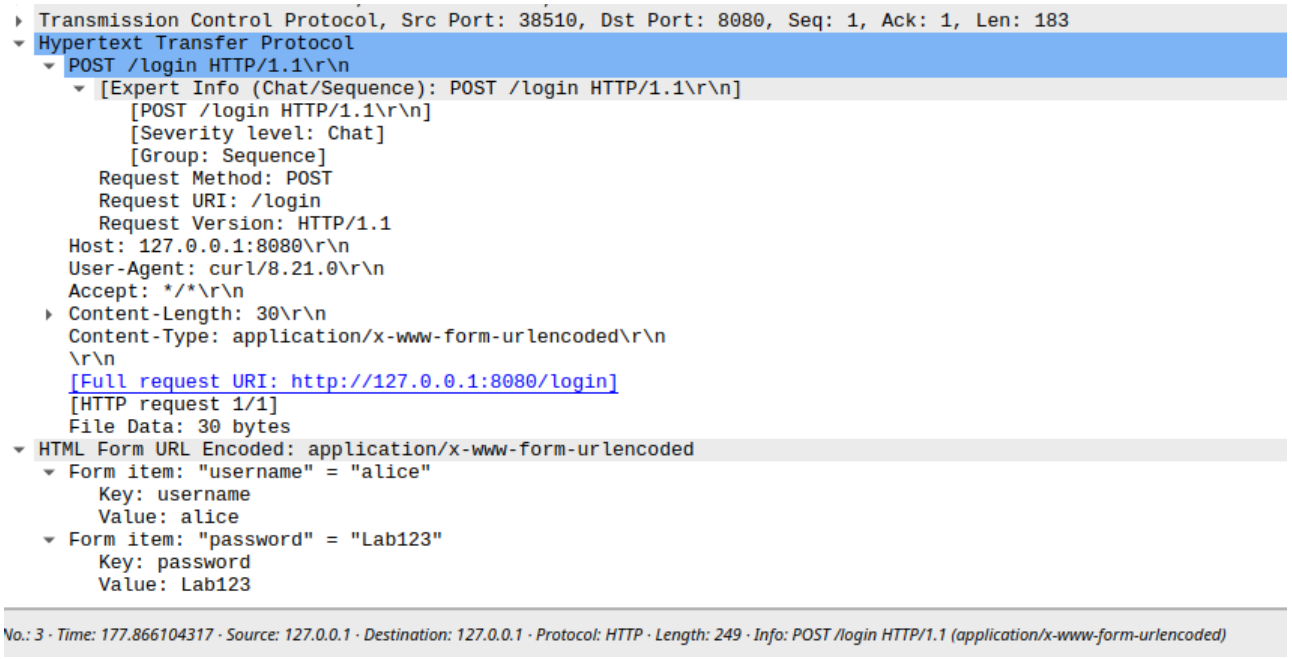
Questions



12. Identify a POST request and record its frame number.

```
Frame No. 3
```

13. Identify the corresponding response and record its frame number.




```
Frame no : 4
```

14. What is the requested URI?

```
Requested URI : /login
```

15. What is the value of the Content-Type field?

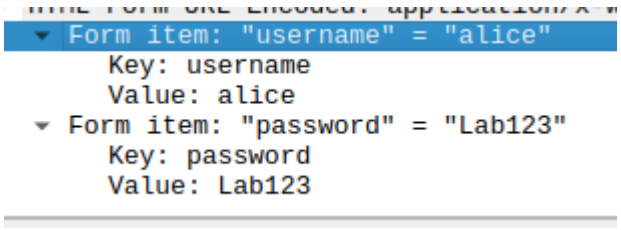
```
Content-Type: application/x-www-form-urlencoded
```

16. Does the request contain a body?

```
Yes
```

17. What information is contained in the request body?

```
username and password from a form
```



18. Identify the Content-Length field, if present.

```
30
```

19. What response status code is returned?

```
POST /login HTTP/1.1
Host: 127.0.0.1:8080
User-Agent: curl/8.21.0
Accept: */*
Content-Length: 30
Content-Type: application/x-www-form-urlencoded

username=alice&password=Lab123<!DOCTYPE HTML>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <style type="text/css">
            :root {
                color-scheme: light dark;
            }
        </style>
        <title>Error response</title>
    </head>
    <body>
        <h1>Error response</h1>
        <p>Error code: 501</p>
        <p>Message: Unsupported method ('POST').</p>
        <p>Error code explanation: 501 - Server does not support this operation.</p>
    </body>
</html>
```

```
501
```

20. What information is contained in the response?

```
<h1>Error response</h1>
<p>Error code: 501</p>
<p>Message: Unsupported method ('POST').</p>
<p>Error code explanation: 501 - Server does not support this
operation.</p>
```

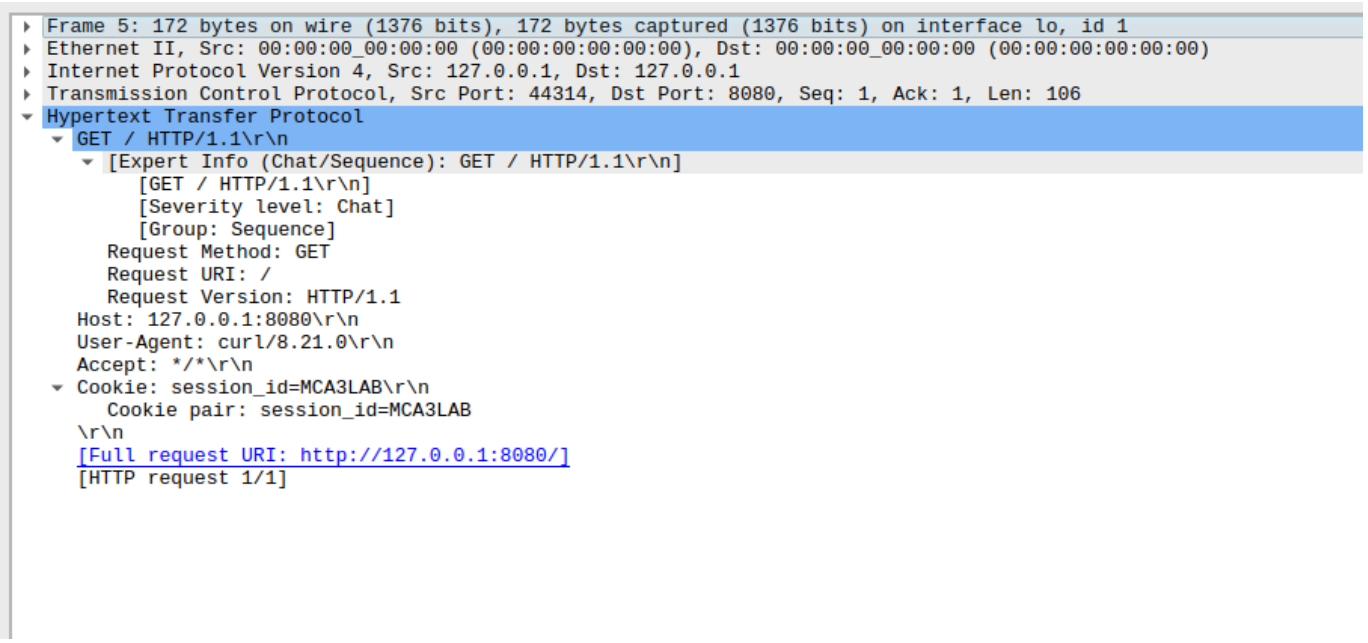
21. Which fields distinguish this POST request from a GET request?

```
method : POST vs GET
Request body : get request doesn't have any but post request do
content-type : only present in post request
content-length : only in post request
form-data : only in post request
```

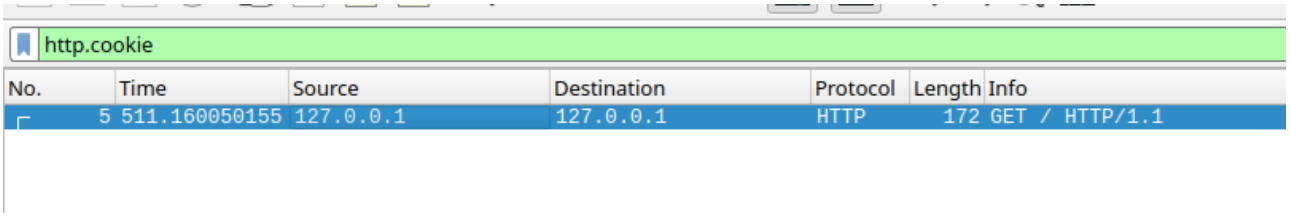
Task 3: Analyze an HTTP Cookie

Identify cookies in the HTTP traffic and determine how they are transmitted between the client and server. (Hint. If http.cookie returns empty frame, use frame 4.) http.cookie

Questions



22. Identify a frame containing a cookie.




```
frame 5
```

23. Record the frame number.

```
Frame 5
```

24. What is the cookie value?

```
Cookie: session_id=MCA3LAB
```

25. Is the cookie transmitted by the client or by the server?

```
Data flow
    Src: 127.0.0.1
    Dst: 127.0.0.1
    Src Port: 44314
    Dst Port: 8080

and packet contains
    GET / HTTP/1.1
    Cookie: session_id=MCA3LAB

So it is from client to server
```

26. In which HTTP header is a client cookie normally transmitted?

```
Cookie: header carries the cookies
```

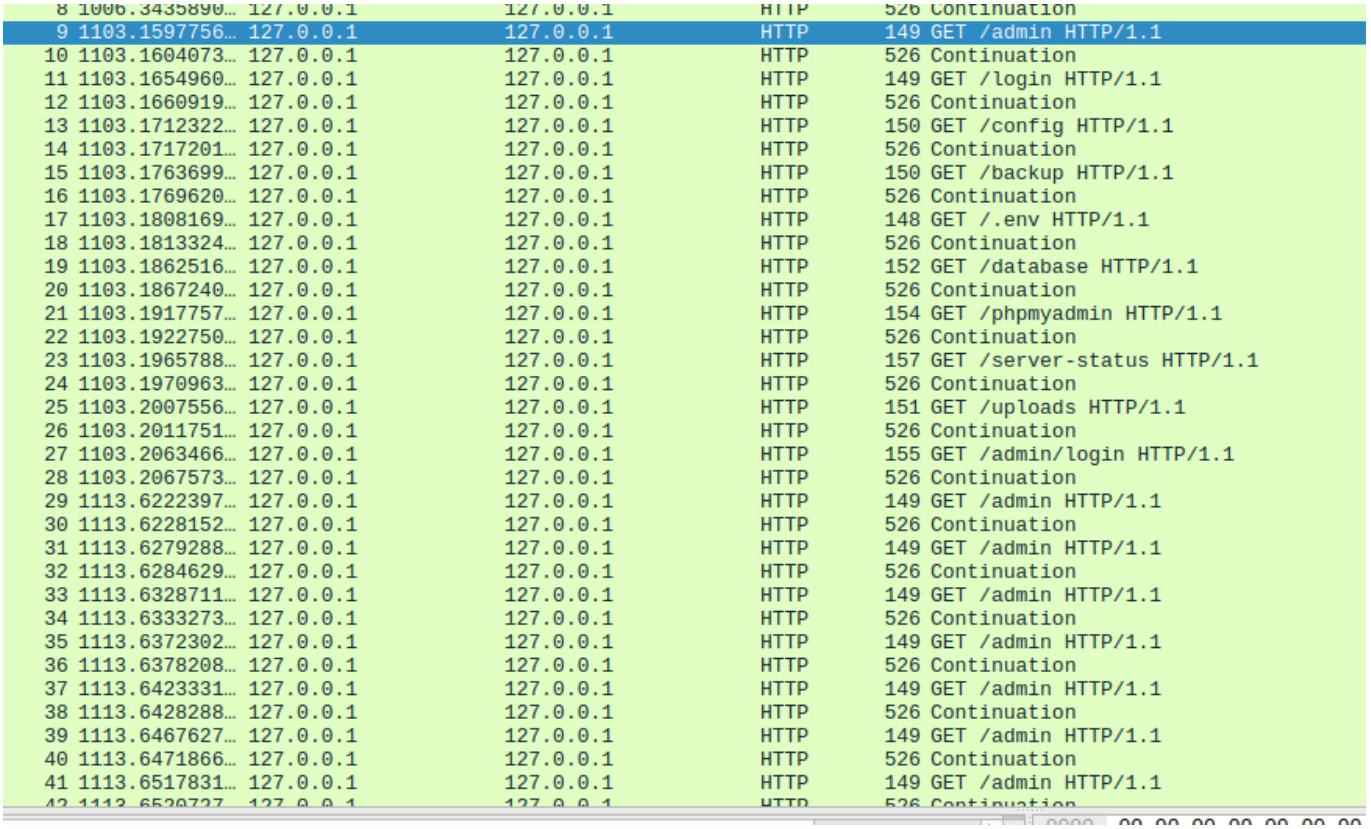
Task 5: Identify Enumeration and Probing Activity

Analyze the HTTP traffic in the provided tracefile.pcapng file and identify packets that may indicate resource enumeration or probing activity. The objective is to distinguish between ordinary HTTP activity and systematic requests that may indicate an attempt to discover or test resources on the server.

5.1 Part A: Identify Resource Enumeration

Examine the HTTP traffic and identify packets that indicate possible resource enumeration. Resource enumeration refers to a sequence of requests intended to discover resources, directories, files, or application endpoints available on the server. (Hint: For this task, students have to examin all the frame and analyze manually by looking at the route (info column) and delta)

Questions



```
 8 1006.3435890… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
 9 1103.1597756… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
10 1103.1604073… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
11 1103.1654960… 127.0.0.1   127.0.0.1   HTTP  149 GET /login HTTP/1.1
12 1103.1660919… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
13 1103.1712322… 127.0.0.1   127.0.0.1   HTTP  150 GET /config HTTP/1.1
14 1103.1717201… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
15 1103.1763699… 127.0.0.1   127.0.0.1   HTTP  150 GET /backup HTTP/1.1
16 1103.1769620… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
17 1103.1808169… 127.0.0.1   127.0.0.1   HTTP  148 GET /.env HTTP/1.1
18 1103.1813324… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
19 1103.1862516… 127.0.0.1   127.0.0.1   HTTP  152 GET /database HTTP/1.1
20 1103.1867240… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
21 1103.1917757… 127.0.0.1   127.0.0.1   HTTP  154 GET /phpmyadmin HTTP/1.1
22 1103.1922750… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
23 1103.1965788… 127.0.0.1   127.0.0.1   HTTP  157 GET /server-status HTTP/1.1
24 1103.1970963… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
25 1103.2007556… 127.0.0.1   127.0.0.1   HTTP  151 GET /uploads HTTP/1.1
26 1103.2011751… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
27 1103.2063466… 127.0.0.1   127.0.0.1   HTTP  155 GET /admin/login HTTP/1.1
28 1103.2067573… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
29 1113.6222397… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
30 1113.6228152… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
31 1113.6279288… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
32 1113.6284629… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
33 1113.6328711… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
34 1113.6333273… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
35 1113.6372302… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
36 1113.6378208… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
37 1113.6423331… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
38 1113.6428288… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
39 1113.6467627… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
40 1113.6471866… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
41 1113.6517831… 127.0.0.1   127.0.0.1   HTTP  149 GET /admin HTTP/1.1
42 1113.6520727… 127.0.0.1   127.0.0.1   HTTP  526 Continuation
```

27. Identify the frame number(s) that you consider indicators of resource enumeration.

```
Frame 9, 11, 13, 15, 17, 19, 21, 23, 25, 27
```

28. Record the requested URI for each identified frame.

Frame no	URI
9	/admin
11	/login
13	/config
15	/backup
17	/.env
19	/database
21	/phpadmin
23	/server-status
25	/uploads
27	/admin/login

29. Are multiple requests made within a short period of time?

```
Yes : 10 requests in almost 0.04657 seconds
```

30. How do the response status codes support your interpretation?

    Not observable for this sequence in the supplied capture.

```
49 → HTTP/1.1 304 Not Modified
50 → HTTP/1.1 304 Not Modified
```

31. Identify the first and last frame belonging to the enumeration sequence.

```
Requests
    First Frame 9
    last Frame 27

Response
    First Frame 10
    last frame 28
```

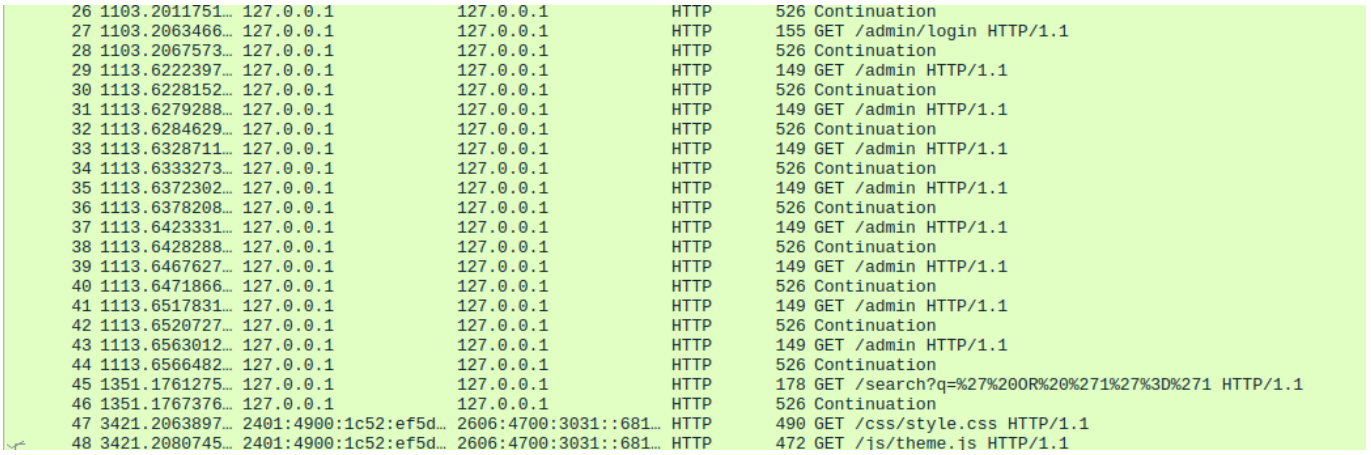
32. Explain why the identified sequence is more consistent with resource enumeration than with ordinary browsing.

```
1. It requests to different potential sensitive routes
2. extreamly short time intervals
3. Response pattern
```

5.2 Part B: Identify Probing Activity

Examine the HTTP traffic and identify packets or sequences that indicate possible probing activity. Probing refers to requests made to test how a particular resource, endpoint, or application function responds.

Questions.



26	1103.2011751…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
27	1103.2063466…	127.0.0.1	127.0.0.1	HTTP	155 GET /admin/login HTTP/1.1
28	1103.2067573…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
29	1113.6222397…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
30	1113.6228152…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
31	1113.6279288…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
32	1113.6284629…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
33	1113.6328711…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
34	1113.6333273…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
35	1113.6372302…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
36	1113.6378208…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
37	1113.6423331…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
38	1113.6428288…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
39	1113.6467627…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
40	1113.6471866…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
41	1113.6517831…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
42	1113.6520727…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
43	1113.6563012…	127.0.0.1	127.0.0.1	HTTP	149 GET /admin HTTP/1.1
44	1113.6566482…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
45	1351.1761275…	127.0.0.1	127.0.0.1	HTTP	178 GET /search?q=%27%20OR%20%271%27%3D%271 HTTP/1.1
46	1351.1767376…	127.0.0.1	127.0.0.1	HTTP	526 Continuation
47	3421.2063897…	2401:4900:1c52:ef5d…	2606:4700:3031::681…	HTTP	490 GET /css/style.css HTTP/1.1
48	3421.2080745…	2401:4900:1c52:ef5d…	2606:4700:3031::681…	HTTP	472 GET /js/theme.js HTTP/1.1

33. Identify the frame number(s) and timestamps that you consider indicators of probing.

```
Frame no 29, 31, 33, 35, 37, 39, 41, 43
```

34. What endpoint is being tested?

```
/admin
```

35. What characteristics distinguish the probing sequence from normal HTTP activity?

```
1. unline different routes it is focused on same /admin route
2. short timespan
3. no normal browser progression
4. the purpose apears to be testing /admin route
```

36. Identify the first and last packet in the probing sequence and provide the frame numbers.

```
First request: Frame 29
Last request: Frame 43
```

Task 6: Identify and Analyze a Suspicious-Looking Frame

Identify an HTTP packet containing suspicious-looking input, extract the relevant parameter, decode any URL-encoded characters, and interpret the result. http.request Inspect the Request URI fields and identify requests containing query parameters. http.request.uri.query If a suspicious parameter has been identified, construct a more specific filter using the relevant URI or parameter value. For example: http.request.uri contains "search"

```
▸ Transmission Control Protocol, Src Port: 49128, Dst Port: 8080, Seq: 1, Ack: 1, Len: 112
▾ Hypertext Transfer Protocol
  ▾ GET /search?q=%27%20OR%20%271%27%3D%271 HTTP/1.1\r\n
    ▾ [Expert Info (Chat/Sequence): GET /search?q=%27%20OR%20%271%27%3D%271 HTTP/1.1\r\n]
        [GET /search?q=%27%20OR%20%271%27%3D%271 HTTP/1.1\r\n]
        [Severity level: Chat]
        [Group: Sequence]
      Request Method: GET
    ▸ Request URI: /search?q=%27%20OR%20%271%27%3D%271
      Request Version: HTTP/1.1
    Host: 127.0.0.1:8080\r\n
    User-Agent: curl/8.21.0\r\n
    Accept: */*\r\n
    \r\n
    [Full request URI: http://127.0.0.1:8080/search?q=%27%20OR%20%271%27%3D%271]
    [HTTP request 1/1]
```

```
GET /search?q=%27%20OR%20%271%27%3D%271 HTTP/1.1
Host: 127.0.0.1:8080
User-Agent: curl/8.21.0
Accept: */*

<!DOCTYPE HTML>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <style type="text/css">
            :root {
                color-scheme: light dark;
            }
        </style>
        <title>Error response</title>
    </head>
    <body>
        <h1>Error response</h1>
        <p>Error code: 404</p>
        <p>Message: File not found.</p>
        <p>Error code explanation: 404 - Nothing matches the given URI.</p>
    </body>
</html>
```

37. Identify one suspicious-looking HTTP request and record the frame.

```
Frame 45
```

38. Identify the query parameter.

```
Request URI: /search?q=%27%20OR%20%271%27%3D%271
```

```
at parameter 'q'
```

39. Identify the encoded value of the parameter.

```
%27%20OR%20%271%27%3D%271
```

40. Determine which characters are URL encoded.

```
%27 : single quote '
%20 : space ' '
%3D : equals sign
```

41. Decode the encoded characters.

```
decodes to ' OR '1'='1' (sql injection command)
```

42. Write the complete decoded parameter value.

```
q = ' OR '1'='1
```

43. What does the decoded value appear to represent?

```
It appears to represent a SQL injection-style input.
```

44. Why might this parameter be considered suspicious?

```
Because a normal search query would generally contain something like
The expression '1'='1' is designed as a condition that evaluates as
true. Combined with the preceding OR, this is characteristic of an
attempt to manipulate an SQL condition.
```